



Advanced Perioperative Nursing in Urology Surgery

This presentation provides a comprehensive overview of advanced perioperative nursing practices in urology surgery. It covers essential topics such as risk assessment, specialized techniques, postoperative care, and management of complications.



by **yananan**

Learning Objectives

1

Analyze and Assess perioperative risks in urology surgery

2

Implement Specialized Techniques to prevent perioperative complications.



Anatomy and Physiology of the Urinary System

Kidneys

The kidneys are vital organs responsible for filtering waste products from the blood and regulating fluid balance.

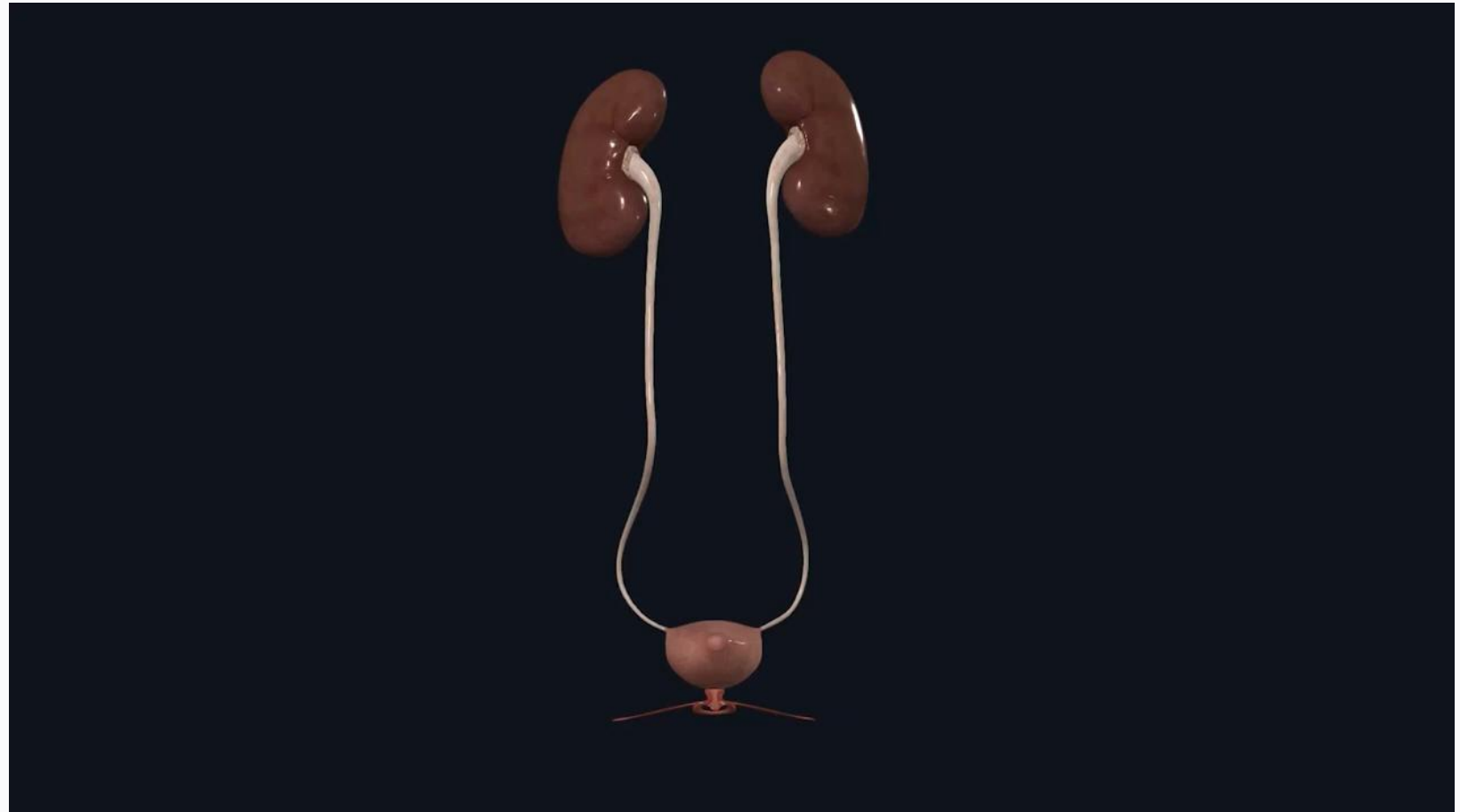
They produce urine, which is transported through the ureters to the bladder for storage.

Urinary Tract

The ureters, bladder, and urethra form the urinary tract, responsible for the transportation and elimination of urine.

Urinary Function

The urinary system plays a crucial role in maintaining homeostasis, controlling blood pressure, and producing hormones like erythropoietin.



Surgical Procedures in Urology

1

Open Surgery

Traditional open surgical techniques involve large incisions : radical nephrectomy, cystectomy, etc.

2

Laparoscopic Surgery

Minimally invasive laparoscopic surgery ; small incisions and a camera to visualize the surgical field, reducing pain and recovery time.

3

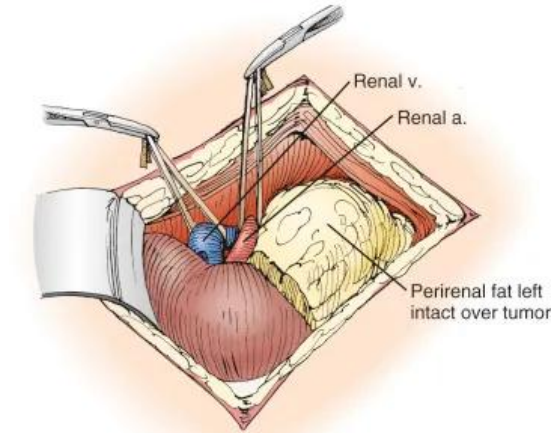
Robotic Surgery

Robotic surgery combines the precision of robotics with the expertise of surgeons, offering enhanced visualization, dexterity, and control during the procedure.

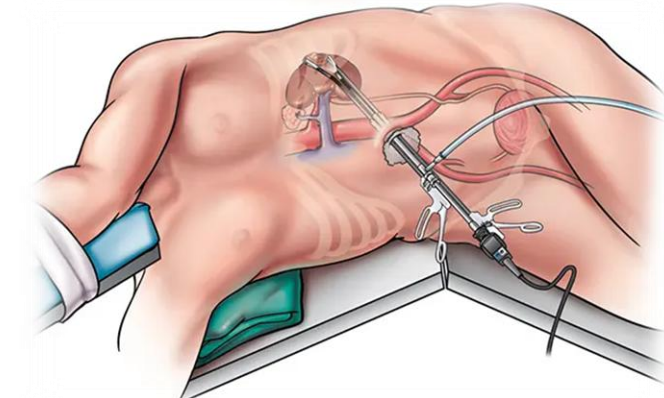
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Endoscopic Surgery

Endoscopic procedures utilize a long, flexible tube with a camera and instruments to access and treat the urinary tract through natural openings, avoiding external incisions.



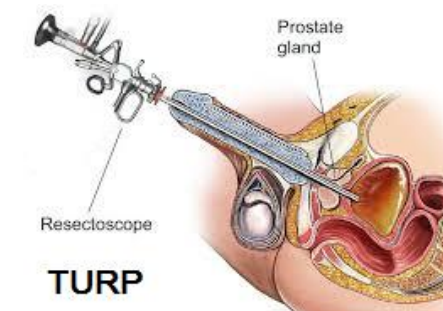
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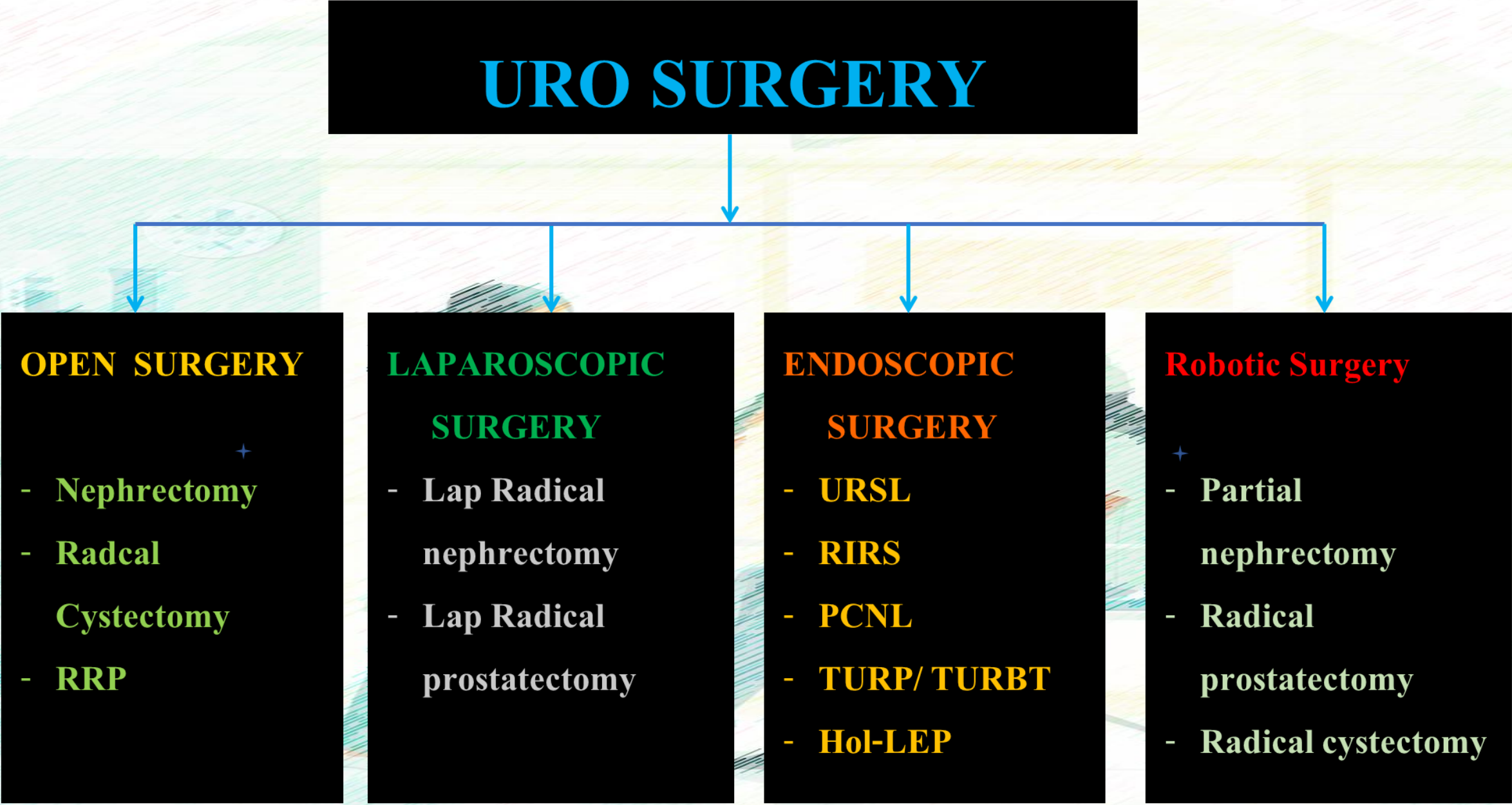
<https://www.eastwesturology.com.au/images/urology/robotic-partial-nephrectomy-1.jpg>



TURP

<https://ccmurology.com/wp-content/uploads/2018/03/TURP-instrument.png>

URO SURGERY



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graph TD; A[URO SURGERY] --> B[OPEN SURGERY]; A --> C[LAPAROSCOPIC SURGERY]; A --> D[ENDOSCOPIC SURGERY]; A --> E[Robotic Surgery];
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OPEN SURGERY

- Nephrectomy
- Radical Cystectomy
- RRP

LAPAROSCOPIC SURGERY

- Lap Radical nephrectomy
- Lap Radical prostatectomy

ENDOSCOPIC SURGERY

- URSL
- RIRS
- PCNL
- TURP/ TURBT
- Hol-LEP

Robotic Surgery

- Partial nephrectomy
- Radical prostatectomy
- Radical cystectomy

Open Surgery in Urology



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Special Precautions

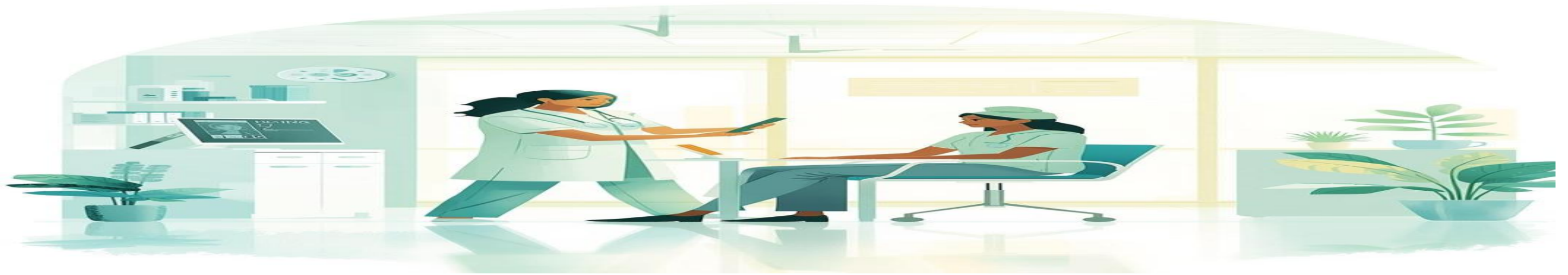
- Maintaining aseptic technique is critical to **prevent infection**.
- The surgical site must be prepared meticulously to **minimize the risk of contamination**.

Management of Blood Loss

- Techniques **for controlling hemorrhage** are essential, including use of hemostatic agents and careful ligation of blood vessels. Blood transfusions may be necessary to manage significant blood loss, and vital signs must be closely monitored.

Infection Control

Prophylactic antibiotics are administered to reduce the risk of infection. Sterile draping and equipment handling are crucial to maintain a sterile field throughout the procedure.



Postoperative Care After Open Surgery

Monitoring for Complications

Nurses monitor for signs of bloodstream infections, such as fever, chills, and increased heart rate.

1

2

Pain Management

pain medication
Patient-controlled analgesia (PCA)

Patient Education

wound care instructions,
including cleaning techniques and signs of infection.

3

Laparoscopic Surgery in Urology

1 Managing Pneumoperitoneum

Nurses are responsible for managing pneumoperitoneum, carefully monitoring intra-abdominal pressure and volume to ensure patient safety.

2 Patient Positioning

Patient positioning is crucial to prevent nerve injuries during laparoscopic procedures. Padding to minimize risks.

3 Continuous Monitoring

Continuous intra-abdominal pressure monitoring is essential to detect potential complications. Capnography is used to assess CO₂ levels and ensure adequate ventilation.





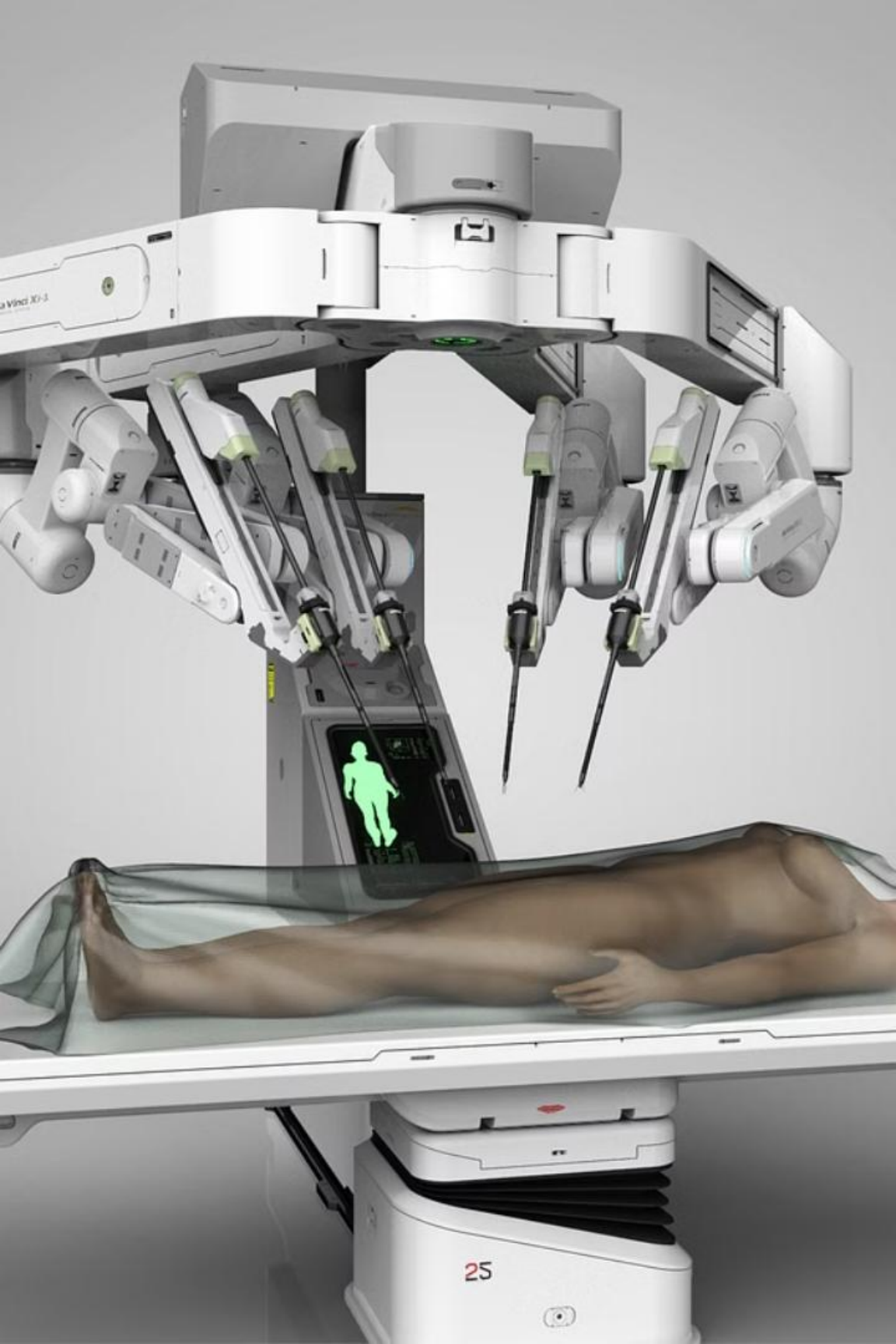
Intraoperative and Postoperative Care in Laparoscopic Surgery

Specialized Instruments

maintaining laparoscopic instruments, ensuring sterility and functionality throughout the procedure.

Managing Complications

gas embolism,
: prevention and treatment.



Robotic Surgery in Urology

1

Control Techniques

Nurses collaborate with the surgical team to understand and manage the interaction between the **robotic system and the instruments.**

2

Risk Assessment

potential risks unique to robotic surgery, such as system **malfunctions and potential complications.**



Intraoperative and Postoperative Care in Robotic Surgery

Managing Technical Issues

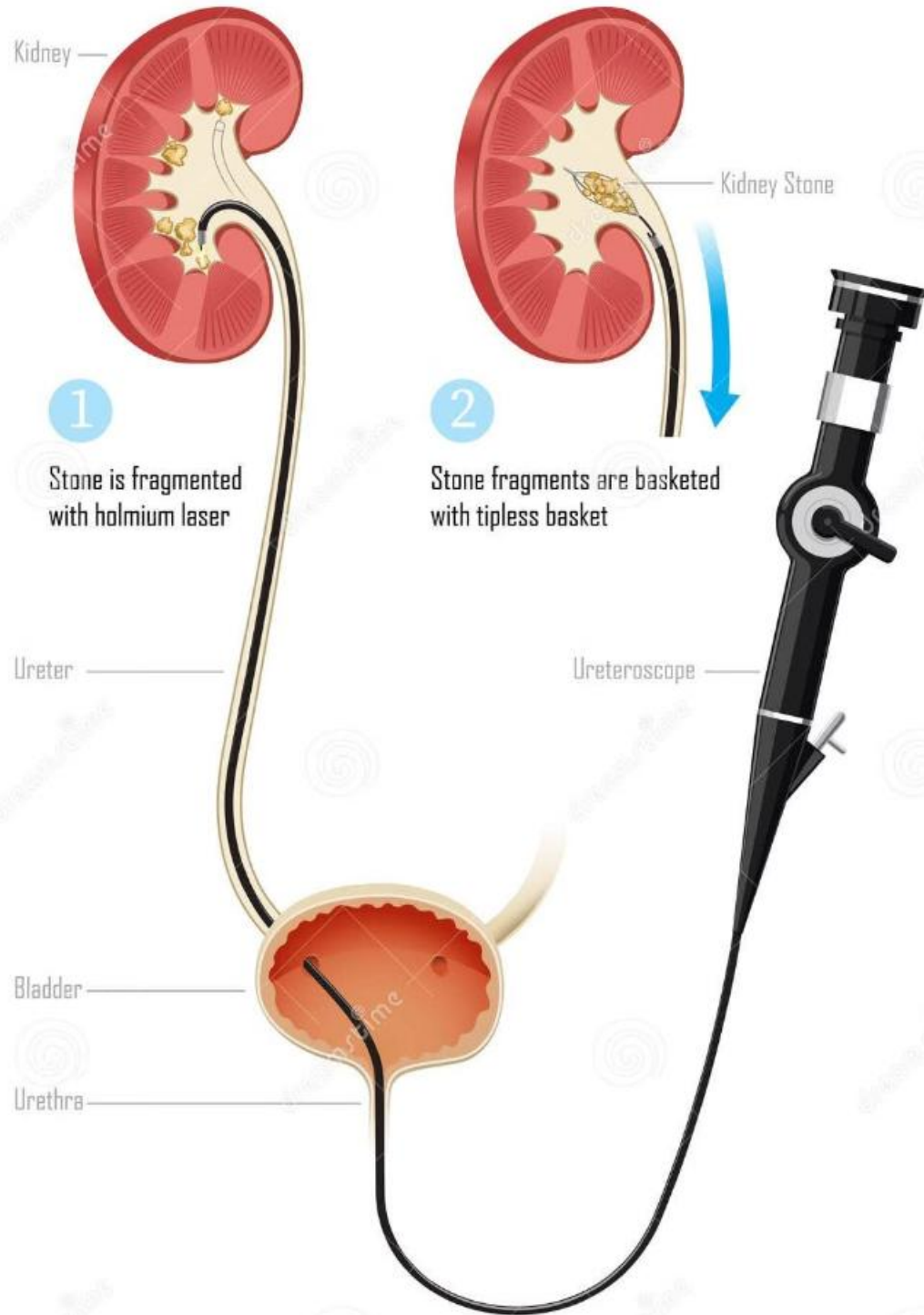
handle robotic system malfunctions and follow established protocols for system failure.

Preventing Complications

Nurses play a critical role in patient safety by ensuring vigilant monitoring throughout the procedure and implementing postoperative protocols to address complications promptly.

Retrograde Intrarenal Surgery (RIRS)

Flexible Ureteroscopy



Endoscopic Surgery in Urology



Infection Prevention

Nurses ensure sterilization of endoscopic instruments to prevent infection. Antimicrobial coatings on instruments provide an additional layer of protection.



Aseptic Techniques

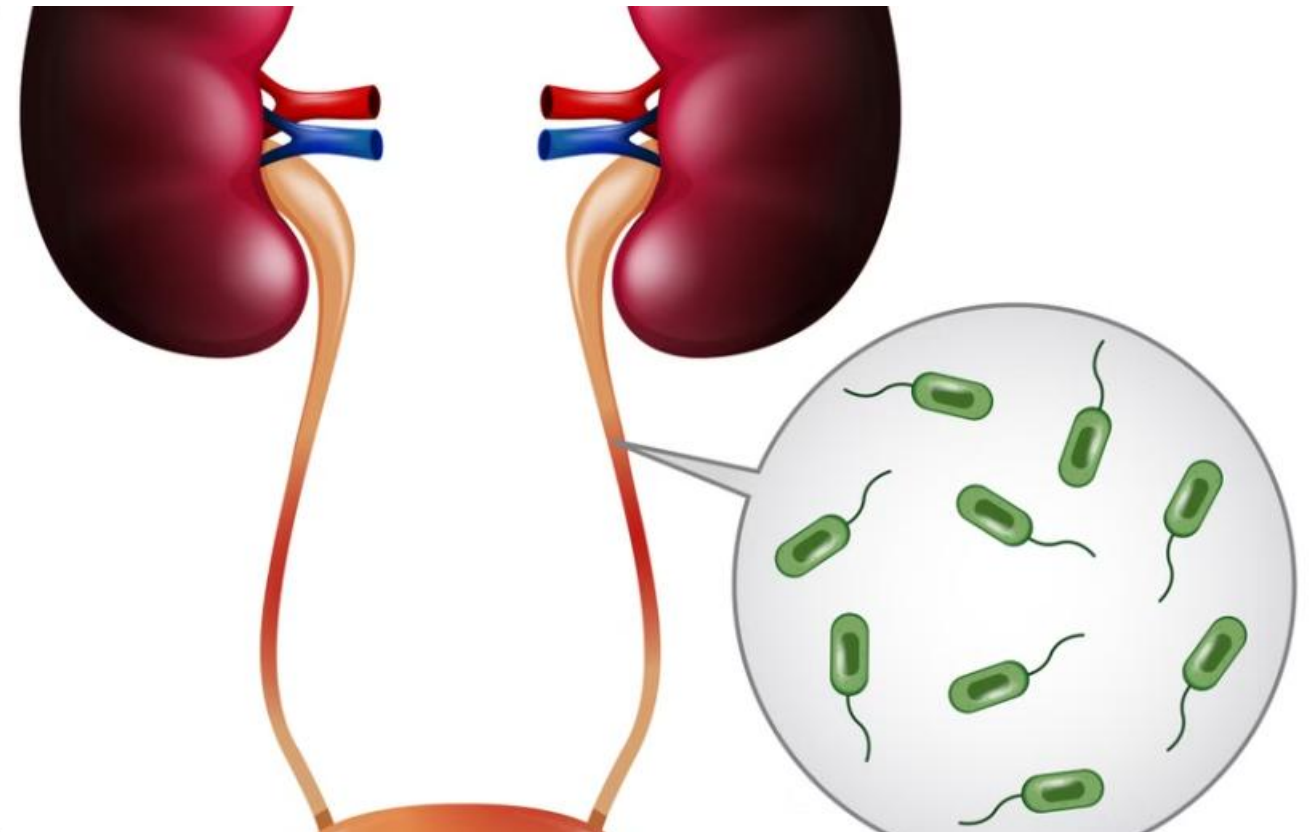
Maintaining a sterile field is crucial during endoscopic procedures. Nurses are responsible for proper handling and disposal of single-use instruments.

Intraoperative and Postoperative Care in Endoscopic Surgery



Monitoring for Complications

Nurses monitor for mucosal injuries or perforations, which are potential complications of endoscopic procedures.



Early Signs of Urinary Tract Infections

Nurses monitor for early signs of urinary tract infections, such as fever, pain, and urinary frequency.

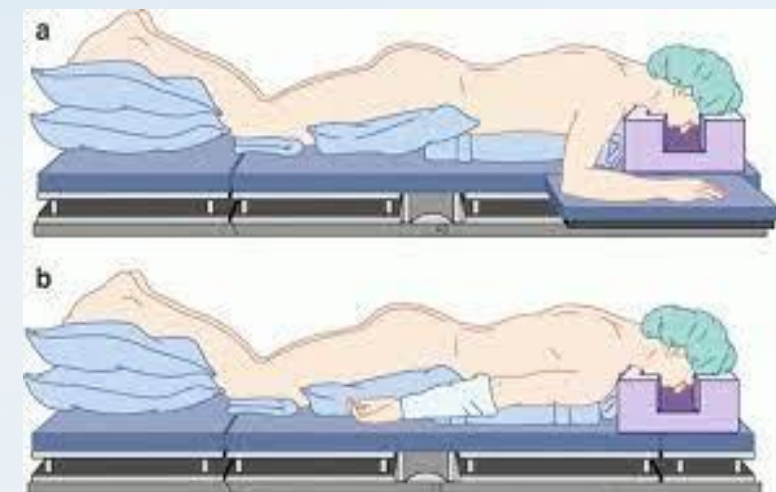
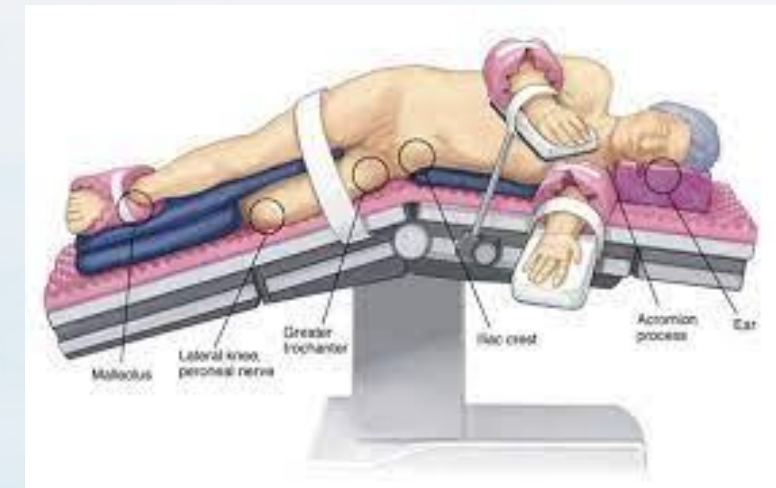
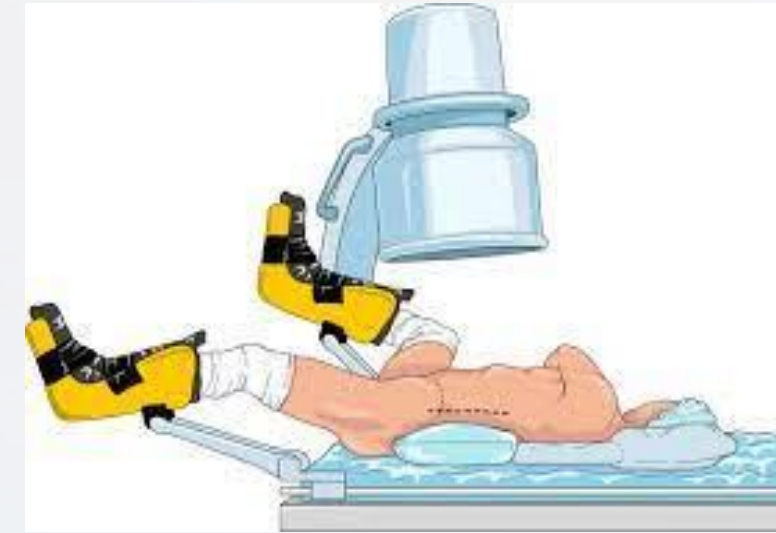
URO

Surgical Positioning



○ Supine

- Lithotomy position
- Lithotomy with Trendelenburg position



○ Lateral decubitus position

○ Prone position



Importance of Proper Positioning

1 Surgical Access

Optimizing positioning allows surgeons to reach the surgical site effectively, facilitating precise procedures.

2 Patient Safety

Positioning minimizes pressure points, protects nerves, and promotes optimal blood flow, safeguarding patient well-being.

3 Comfort and Safety

Proper positioning ensures comfort, prevents complications, and ensures a safe and efficient surgical environment.

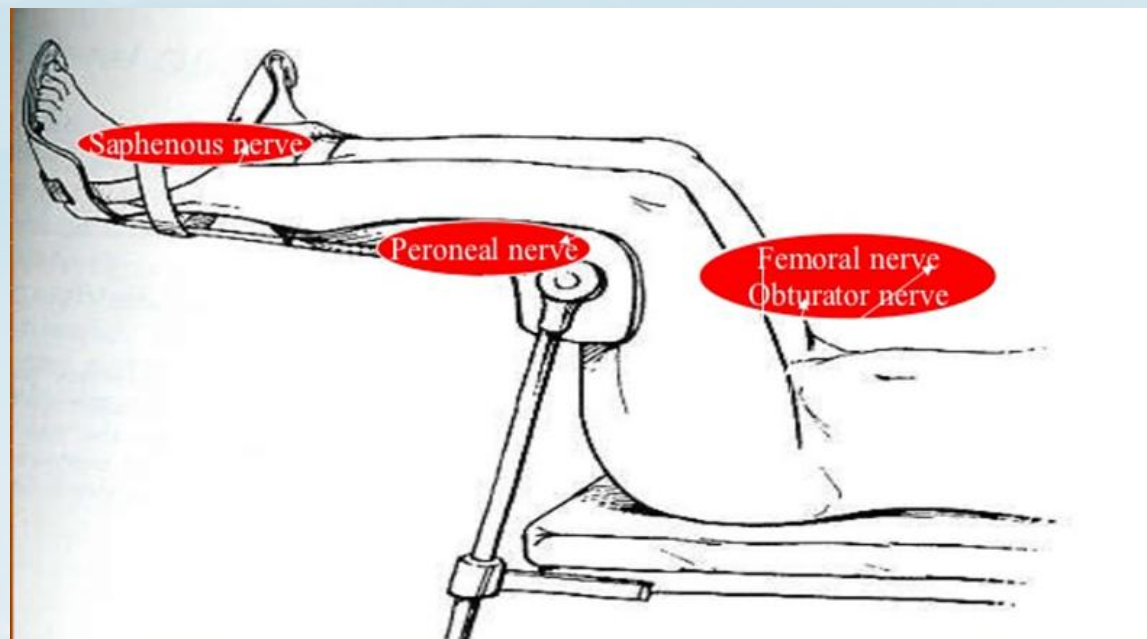
4 Sterility

Maintaining sterility during positioning helps minimize the risk of infection and ensures the safety of both the patient and surgical team.

Lithotomy Position for TUR, URS, RIRS

Common Complications

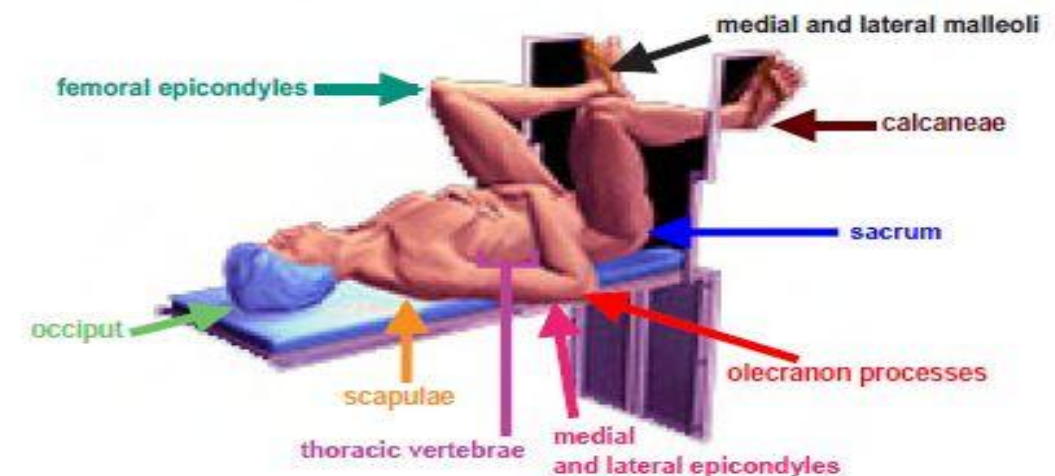
Nerve injuries, like peroneal nerve damage, can occur due to pressure or stretching. Circulatory issues, such as decreased blood flow to the legs, are also potential risks.

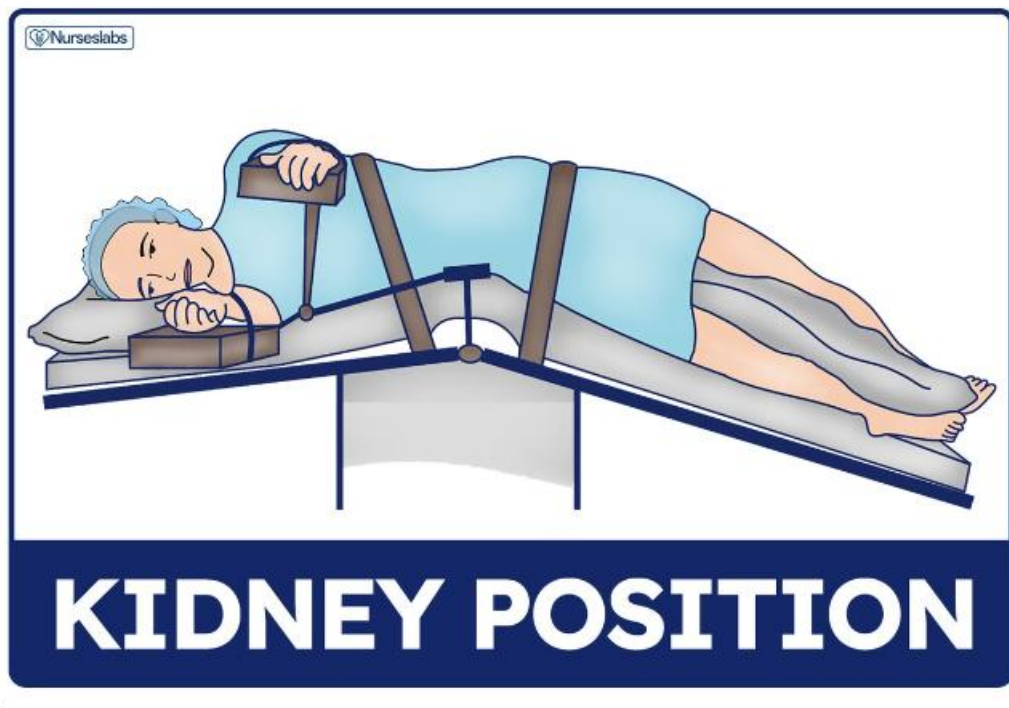


Prevention Strategies

Proper padding and positioning, especially in the lower extremities and hips, are crucial. Regular repositioning helps alleviate pressure and promote circulation.

Figure 6: Pressure points in the lithotomy position





Lateral or Kidney Position for Nephrectomy

Respiratory Compromise

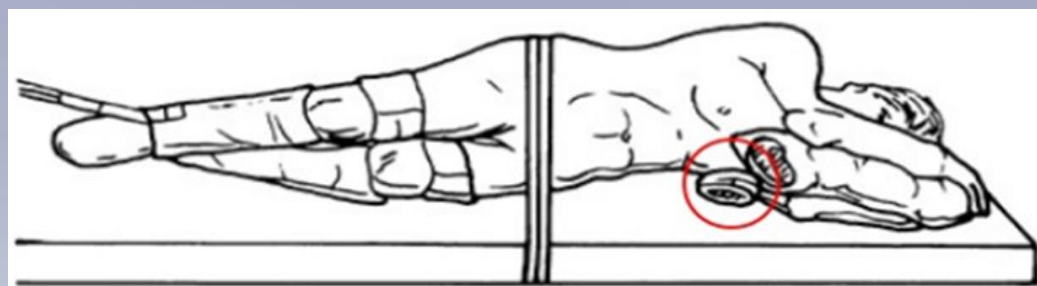
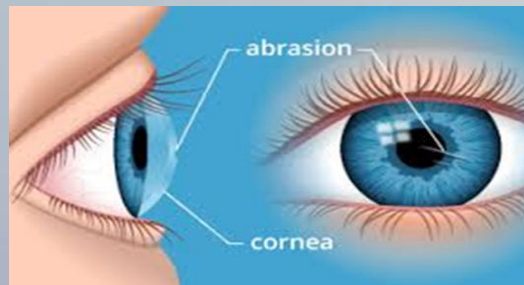
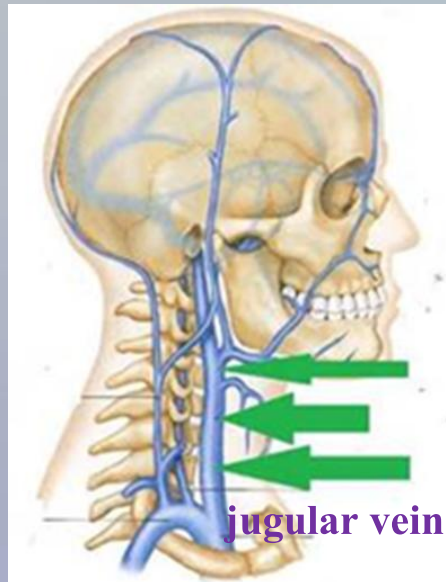
The lateral position can **restrict lung expansion**, potentially leading to respiratory difficulties. Close monitoring is essential.

Nerve Injuries

Pressure on the **ulnar nerve in the arm** can cause nerve damage, leading to numbness or tingling. Proper positioning and padding are crucial.

Prevention Techniques

Securing the limbs to prevent movement during surgery is vital. Continuous monitoring of respiratory status helps detect and address any issues promptly.



Prone Position for PCNL

1

Pressure Ulcers

Prolonged pressure on the chest and abdomen can lead to skin breakdown and ulcer formation. Regular repositioning and pressure-relieving devices are vital.

2

Ocular Complications

Pressure on the eyes can cause corneal abrasions or other eye injuries. Eye protection is essential during prone positioning.

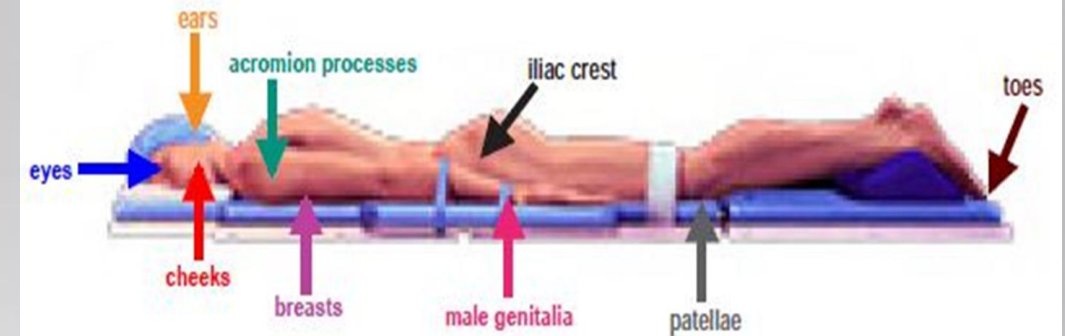
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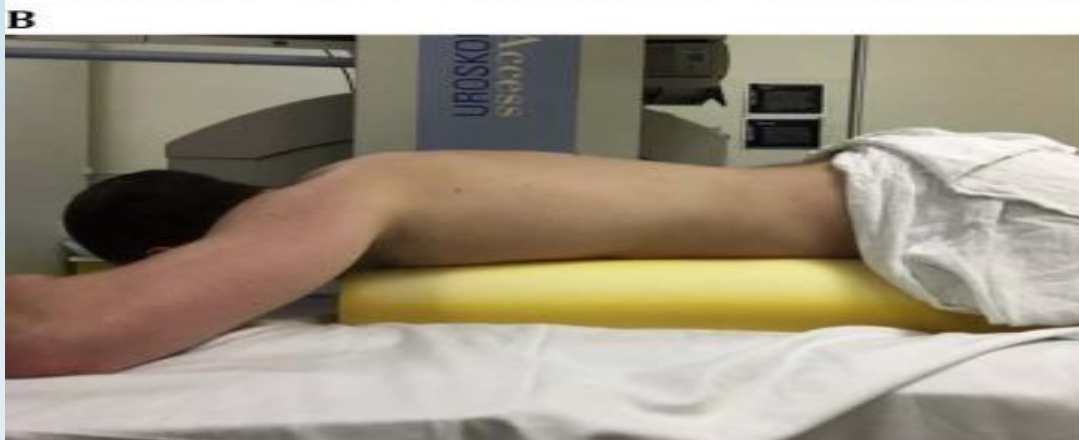
Prevention Strategies

Pressure-relieving devices, such as gel pads or pillows, help distribute pressure and reduce the risk of ulcers. Regular monitoring of eye protection ensures its effectiveness.



Figure 10: Pressure points in prone position





Monitoring and Adjusting Positioning

1

Initial Assessment

Thorough assessment of the patient's physical condition and specific needs before positioning is essential.

2

Continuous Monitoring

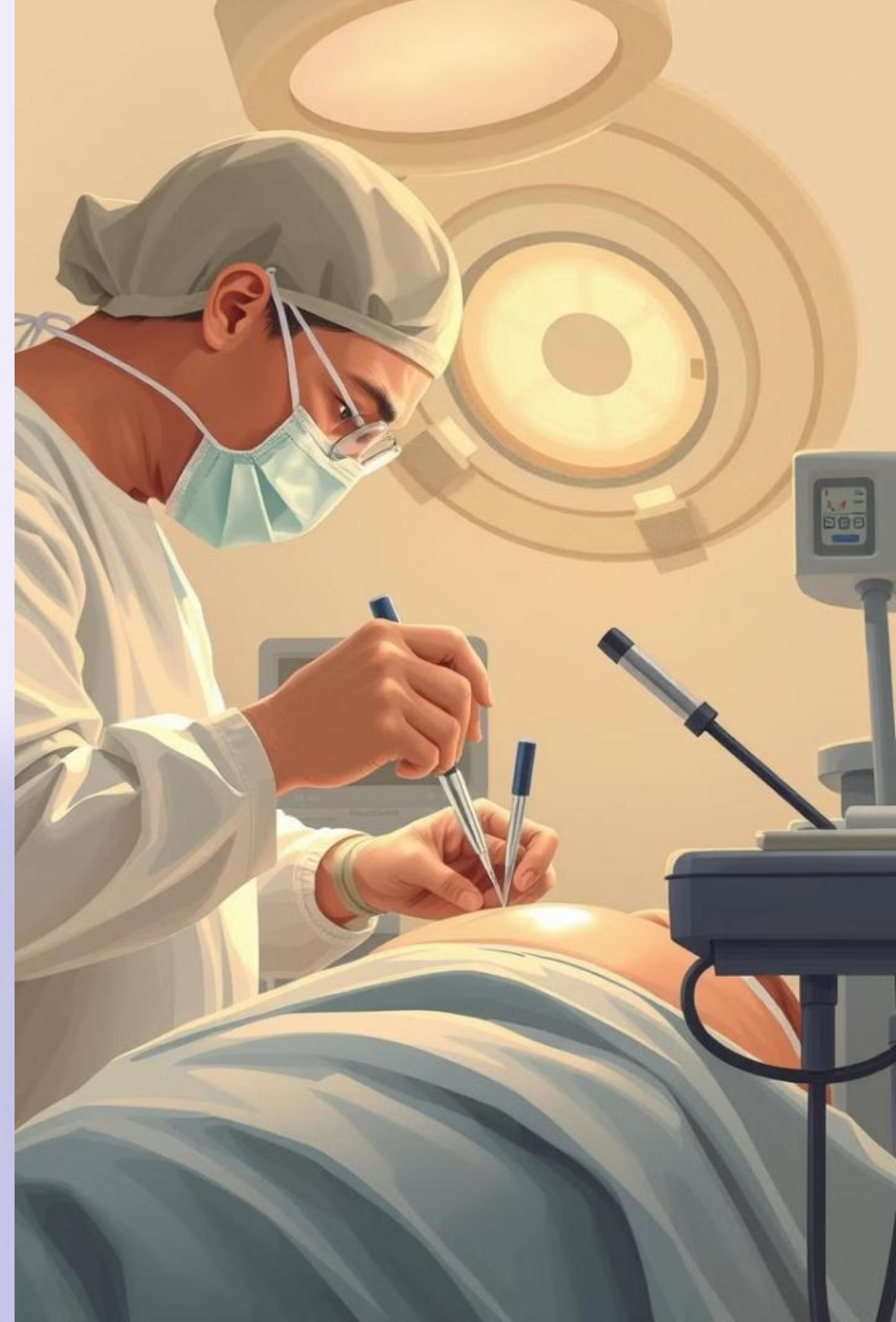
Regular checks for **pressure points, nerve compression**, and circulation are crucial throughout the surgery.

3

Adjustments as Needed

Adjusting positioning based on patient comfort, vital signs, and surgical needs ensures optimal safety and comfort.

Perioperative Care in Urology Surgery



Uro Sepsis

1 Risk Assessment

Uro sepsis is a serious infection infection that can occur after after urological procedures. Identifying patients at high risk risk is crucial for early intervention.

3 Laboratory Markers

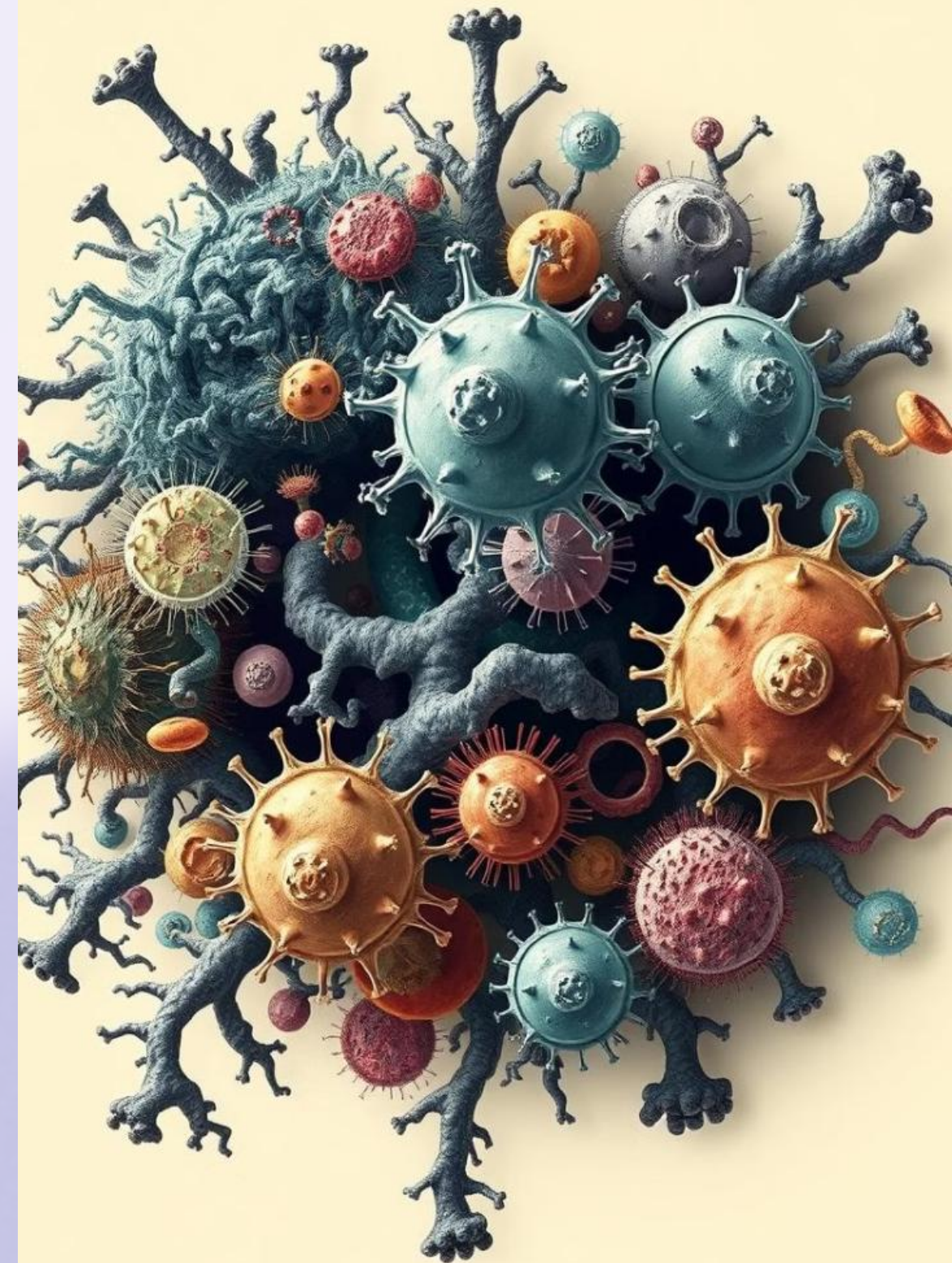
Laboratory tests, such as blood blood cultures, can confirm the the presence of infection and and guide antibiotic treatment. treatment.

2 Monitoring Vital Signs

Close monitoring of vital signs, signs, such as temperature, heart rate, and blood pressure, pressure, can help detect early early signs of infection.

4 Surveillance Strategies Strategies

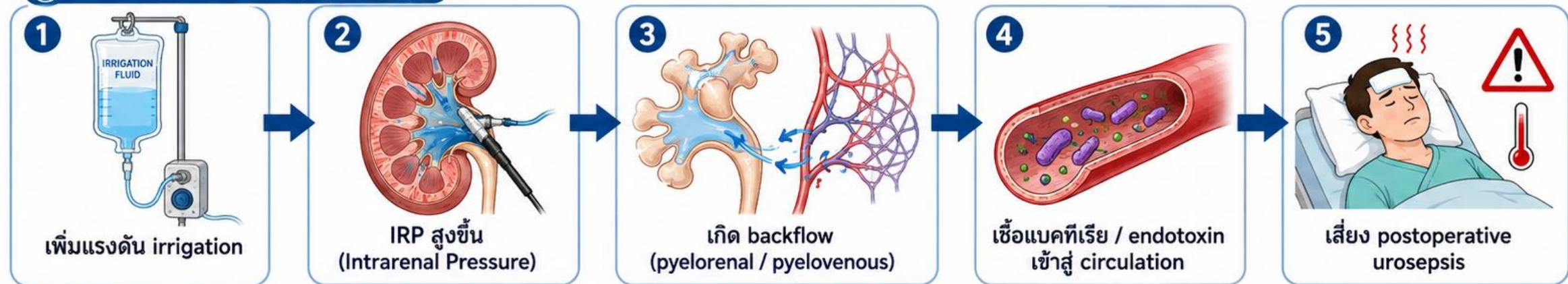
Regular patient assessments assessments and early detection detection protocols are essential essential for timely intervention intervention and reducing the the risk of complications.



Irrigation → IRP → Backflow → Urosepsis

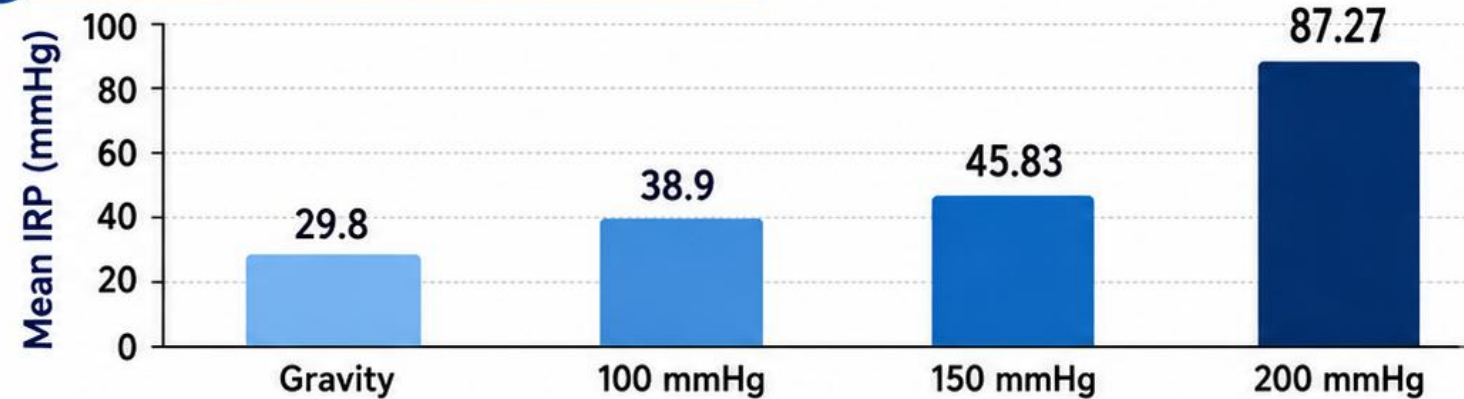
สรุปจากงานวิจัย Croghan et al., BJU International 2023

① กลไกเชื่อมโยง (Pathophysiology)



★ **ประเด็นสำคัญ:** Irrigation ไม่ได้มีผลแก่การมองเห็น แต่มีผลต่อความปลอดภัยของผู้ป่วยด้วย

② Mean IRP ระหว่าง FURS with 11/13-F UAS



ยิ่งเพิ่มแรงดัน irrigation → IRP ยิ่งสูง

ผู้ป่วยที่เกิด urosepsis มี IRP ช่วง FURS สูงกว่า control **81.7 vs 38.53 mmHg**



สรุป: IRP ที่สูงและผันผวนสัมพันธ์กับภาวะแทรกซ้อน โดยเฉพาะใช้หลังผ่าตัดและ urosepsis

Croghan, S.M., Cunnane, E.M., O'Meara, S., Muheilan, M., Cunnane, C.V., Patterson, K., Skolarikos, A., Somani, B., Jack, G.S., Forde, J.C., O'Brien, F.J., Walsh, M.T., Manecksha, R.P., McGuire, B.B. and Davis, N.F. (2023), In vivo ureteroscopic intrarenal pressures and clinical outcomes: a multi-institutional analysis of 120 consecutive patients. BJU Int, 132: 531-540. <https://doi.org/10.1111/bju.16169>

Prevention of Uro Sepsis

Aseptic Techniques

Strict adherence to sterilization protocols is paramount. All surgical instruments and equipment must be properly sterilized to minimize the risk of infection.

Antibiotic Management

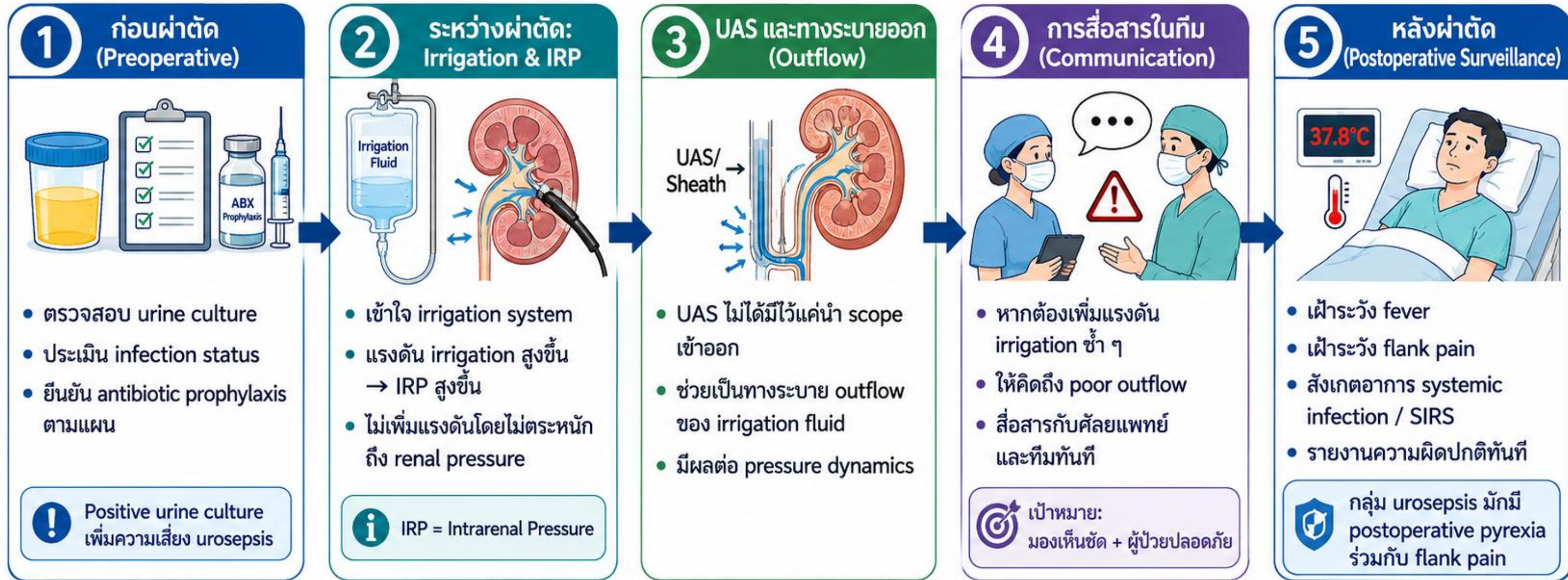
Prophylactic antibiotics are often used to prevent infections. Appropriate selection and administration of antibiotics are essential, along with monitoring for antibiotic resistance.

Hand Hygiene

Proper hand hygiene is fundamental. Surgical personnel must thoroughly wash their hands before and after procedures to reduce the risk of bacterial transmission.

บทบาทพยาบาลห้องผ่าตัดในการลดความเสี่ยง Urosepsis ระหว่าง URS/FURS

Perioperative nursing focus: ก่อนผ่าตัด – ระหว่างผ่าตัด – หลังผ่าตัด



Key message: Irrigation ไม่ได้มีผลแค่การมองเห็น แต่มีผลต่อความปลอดภัยของผู้ป่วยด้วย



สรุป: พยาบาลห้องผ่าตัดมีบทบาทสำคัญในการประเมิน ป้องกัน ฝ้าระวัง และสื่อสาร เพื่อลดความเสี่ยง urosepsis

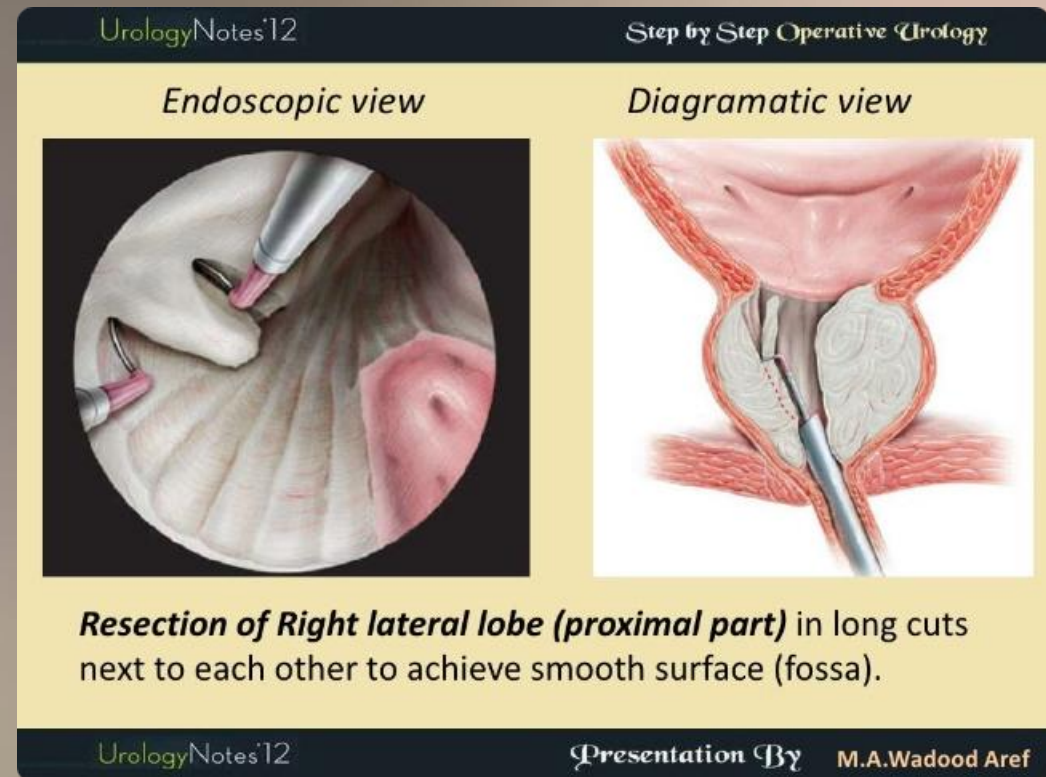
TURP Syndrome

Causes

Absorption of irrigation fluids during transurethral resection of the prostate (TURP) surgery can lead to fluid overload and electrolyte imbalances.

Symptoms

Hyponatremia, confusion, hypertension/hypotension, visual disturbances, seizures.





Preventing TURP Syndrome

Fluid Management

Limiting the absorption of irrigation fluids is essential. Surgeons use techniques to minimize fluid absorption, such as using isotonic solutions when possible.

Electrolyte Monitoring

Regular electrolyte checks throughout the procedure are crucial to identify and correct any imbalances. Close monitoring of fluid balance is essential to prevent complications.

Hypothermia in Surgery

1

Monitoring Body Temperature

Continuous temperature monitoring during surgery is crucial to identify early signs of hypothermia.

2

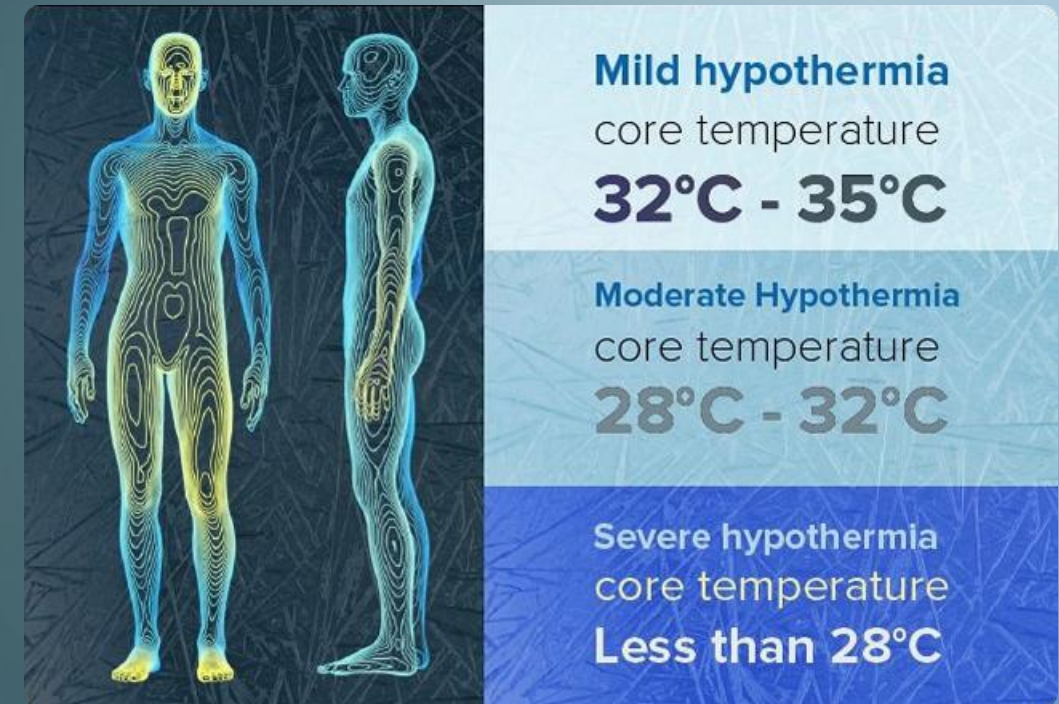
Recognizing Signs

Signs of hypothermia include shivering, decreased heart rate, and rate, and low blood pressure.

3

Impact on Patient Outcomes

Hypothermia can increase the risk of infection, delay wound wound healing, and prolong recovery time.



Prevention of Hypothermia



Warming Techniques

Forced-air warming blankets and heated intravenous fluids can help maintain a safe body temperature.



Environmental Controls

Maintaining a comfortable operating room temperature and minimizing exposure of the patient's body can reduce the risk of hypothermia.



VTE in Urology Surgery

Venous thromboembolism (VTE) is a serious complication that can occur after urological surgery. This presentation will discuss the epidemiology, risk factors, prophylaxis strategies, and management of VTE in urology patients.



Epidemiology of VTE in Urology Patients

1

Risk Factors

Factors that increase the risk of VTE include age, obesity, smoking, and cancer.

2

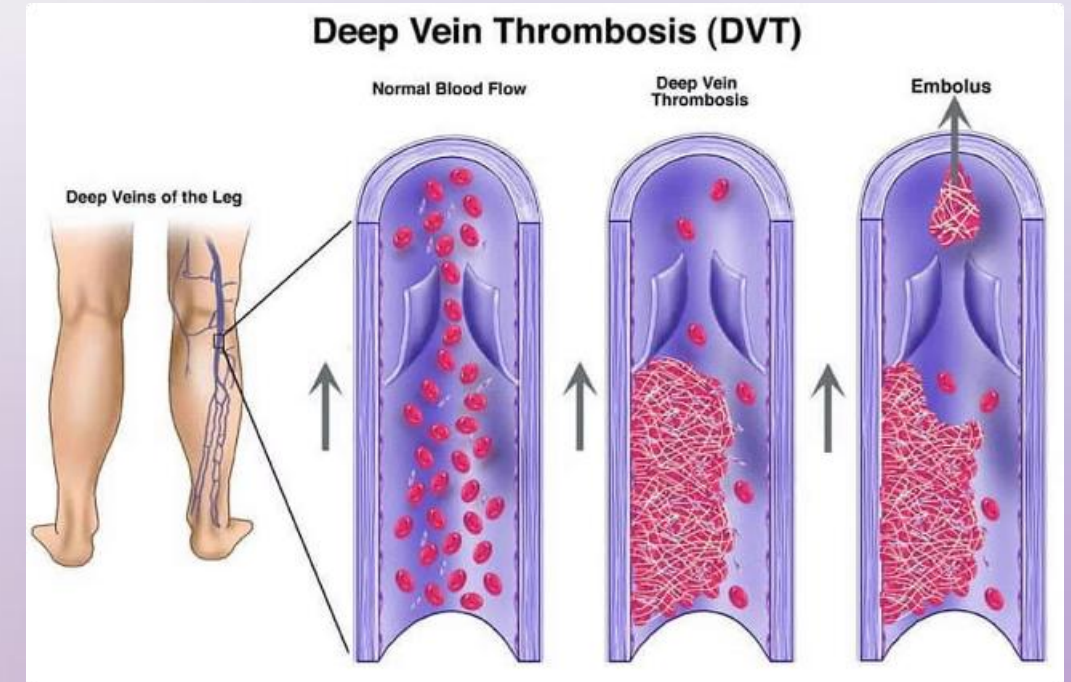
Incidence

The incidence of VTE after urology surgery is estimated to be between 1% and 10%.

3

Consequences

VTE can lead to serious complications such as pulmonary embolism (PE) and even death.



Risk Factors for VTE in Urology Procedures

Patient-Related

These include age, obesity, smoking, cancer, and prior history of VTE.

Procedure-Related

The type of surgery performed, its duration, and the extent of tissue manipulation are important considerations.

Medication-Related

Certain medications, such as hormonal therapy and chemotherapy, can increase the risk of VTE.



IPC can

Reduce the chance of asymptomatic and distal DVT by 60% and reduce the chance of proximal DVT by 50%.

Roderick P, Ferris G, Wilson K, et al. T; 2005

Mechanical Prophylaxis Modalities

Mechanical prophylaxis methods are often used in conjunction with pharmacological therapy to further reduce the risk of VTE.

Compression Stockings

These elastic stockings exert pressure on the legs to promote blood flow and reduce the risk of DVT.

Intermittent Pneumatic Compression (IPC)

IPC devices use air pressure to inflate and deflate sleeves wrapped around the legs, improving blood circulation.

Venous Thromboembolism (VTE) Filters

Filters placed in the inferior vena cava can prevent emboli from traveling to the lungs.



Perioperative VTE Monitoring and Management

Close monitoring and timely management are essential for identifying and treating VTE in urology patients.

1

Clinical Assessment

Regular assessment for signs and symptoms of DVT, such as leg swelling, pain, and redness.

2

Imaging Studies

Ultrasound or venography may be used to diagnose DVT.

3

Anticoagulation Therapy

Treatment typically involves anticoagulation with heparin, warfarin, or newer oral anticoagulants.

4

VTE Prevention

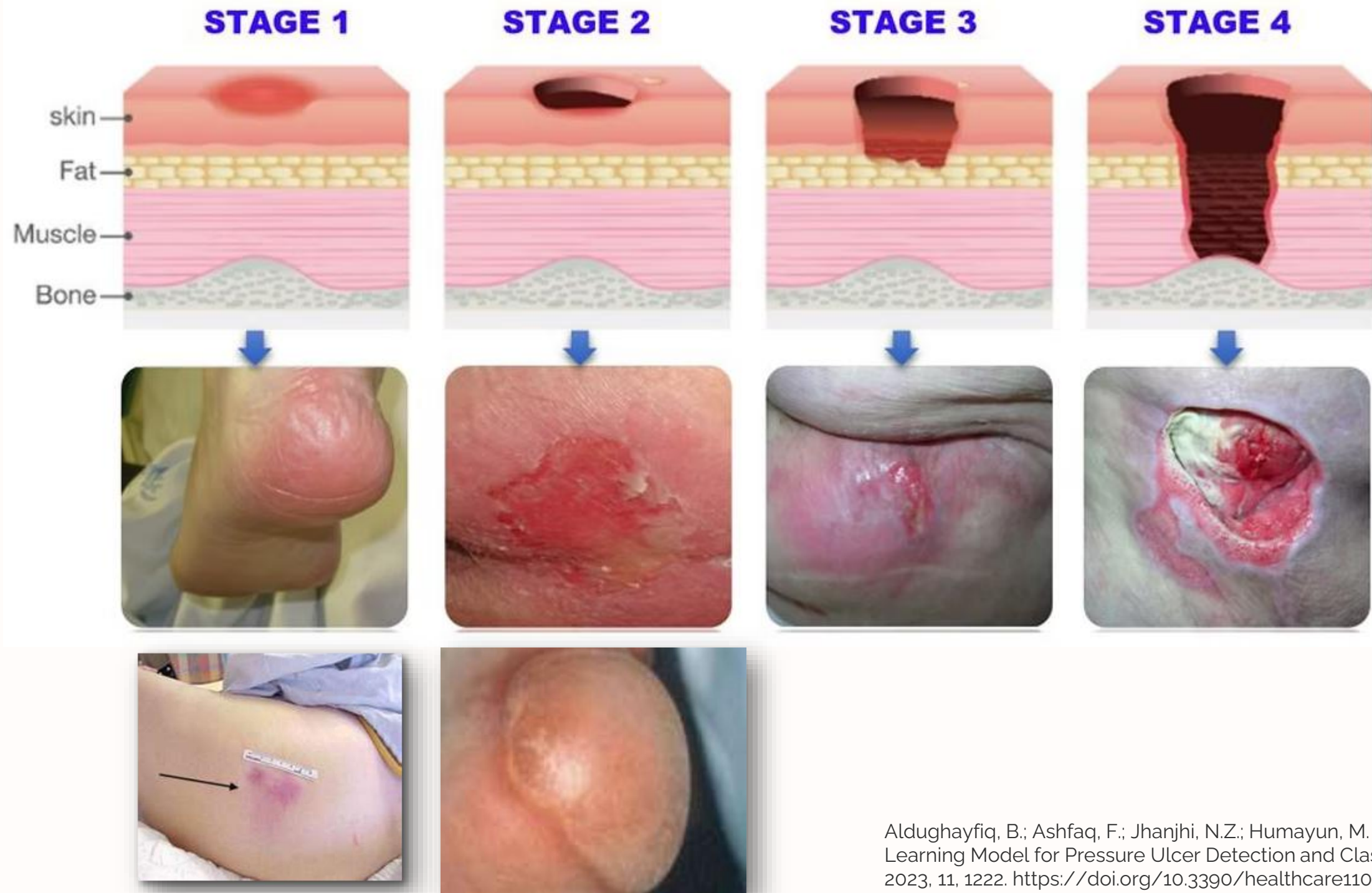
Maintaining adequate hydration, early ambulation, and compression stockings are crucial in preventing further VTE.

Preventing Pressure Injuries in Perioperative Urology Nursing



Patient Safety Innovations





Aldughayfiq, B.; Ashfaq, F.; Jhanjhi, N.Z.; Humayun, M. YOLO-Based Deep Learning Model for Pressure Ulcer Detection and Classification. Healthcare 2023, 11, 1222. <https://doi.org/10.3390/healthcare11091222>

Risk Factors for Pressure Injuries

1 Prolonged Immobility

Surgical procedures often involve prolonged immobility, making patients more susceptible to pressure injuries.

2 Inadequate Padding and Support

Insufficient padding and support during surgery can contribute to the development of pressure injuries.

3 Compromised Circulation

Patients with underlying conditions affecting blood flow, like diabetes or peripheral artery disease, are more prone to pressure injuries.

4 Skin Conditions

Pre-existing skin conditions, such as eczema or psoriasis, can make the skin more fragile and vulnerable to injury.



Perioperative Pressure Ulcer Prevention

Scott Trigger s

- Age **>62**
- Albumin **<3.5**
- ASA Score **≥3**
- Time on the table **≥3 hr**

Score 2 or more = HIGH RISK

At Risk Protocol

- Use high specification OR table pads
- Offload heels
- Use special padding for high risk body areas
- Use only approved positioning devices
- Use boot type stirrups
- Post op: Early movement, skin assessment, and pressure management

Prevention Techniques for Pressure Injuries

1

Regular Repositioning

Frequent repositioning of the patient during surgery helps to alleviate pressure on sensitive areas, reducing the risk of injury.

2

Pressure-Relieving Devices

Special mattresses, cushions, and padding can redistribute pressure and prevent the development of pressure injuries.

3

Skin Care

Proper skin care, including gentle cleansing and moisturizing, helps maintain the skin's integrity and reduce the risk of injury.

Positioning Process

- Collaborative process

- Selection of equipment
- Preoperative assessment
- Positioning
- Documentation
- Postoperative evaluation





Managing Pneumoperitoneum in Laparoscopic Surgery

Physiological Impacts

1. Increased intra-abdominal pressure can affect respiration and circulation.
2. Potential for gas embolism.

Monitoring During Surgery

1. Continuous monitoring of intra-abdominal pressure to ensure optimal levels.
2. Assess the patient's respiratory and cardiovascular status through vital signs and ECG monitoring.

Pressure Management Techniques

1

Optimal Intra-abdominal Pressure

Maintaining an optimal intra-abdominal pressure during laparoscopic surgery is essential to minimize complications.

2

Insufflation Rate Adjustments

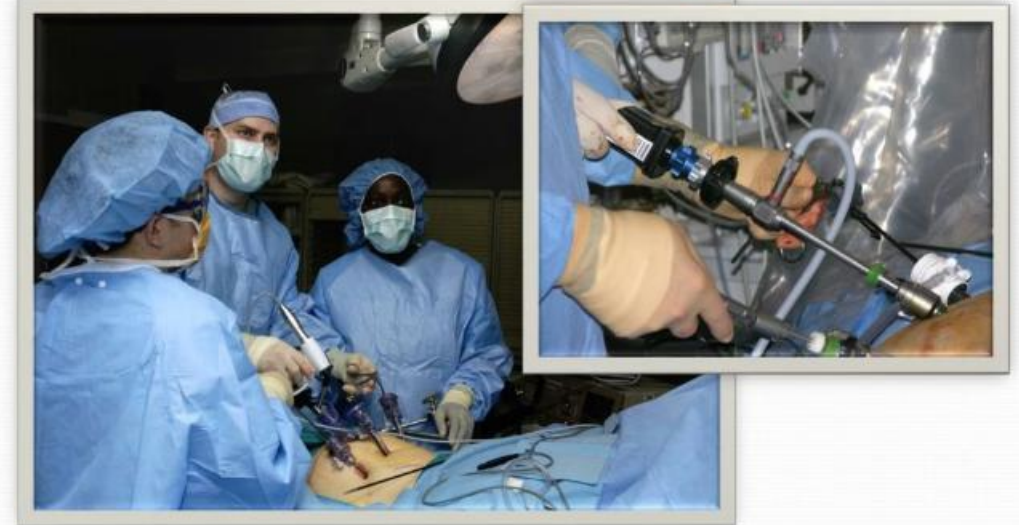
Adjusting the insufflation rate, which controls the flow of gas into the abdominal cavity, can help maintain optimal pressure.

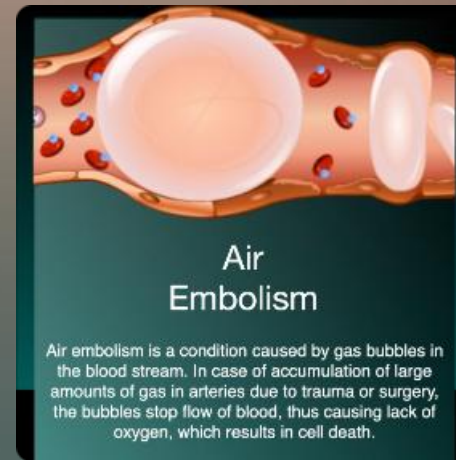
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Fluid Management

Maintaining adequate hydration and managing fluid levels during surgery is crucial for stabilizing intra-abdominal pressure.

Laparoscopic Surgery





Preventive Measures for Pneumoperitoneum Complications

Appropriate Gas Types

Using inert gases like carbon dioxide, helps minimize the risk of complications.

Flow Rate Control

Maintaining a controlled flow rate of gas into the abdominal cavity prevents excessive pressure buildup and potential complications.

Gas Embolism Protocols

Having established protocols for managing gas embolism, which occurs when gas bubbles enter the bloodstream, is crucial for patient safety.

CO2 subcutaneous emphysema

It is caused by accidentally or intentionally inserting CO2 outside the abdomen, such as kidney surgery, lymph node removal, etc.

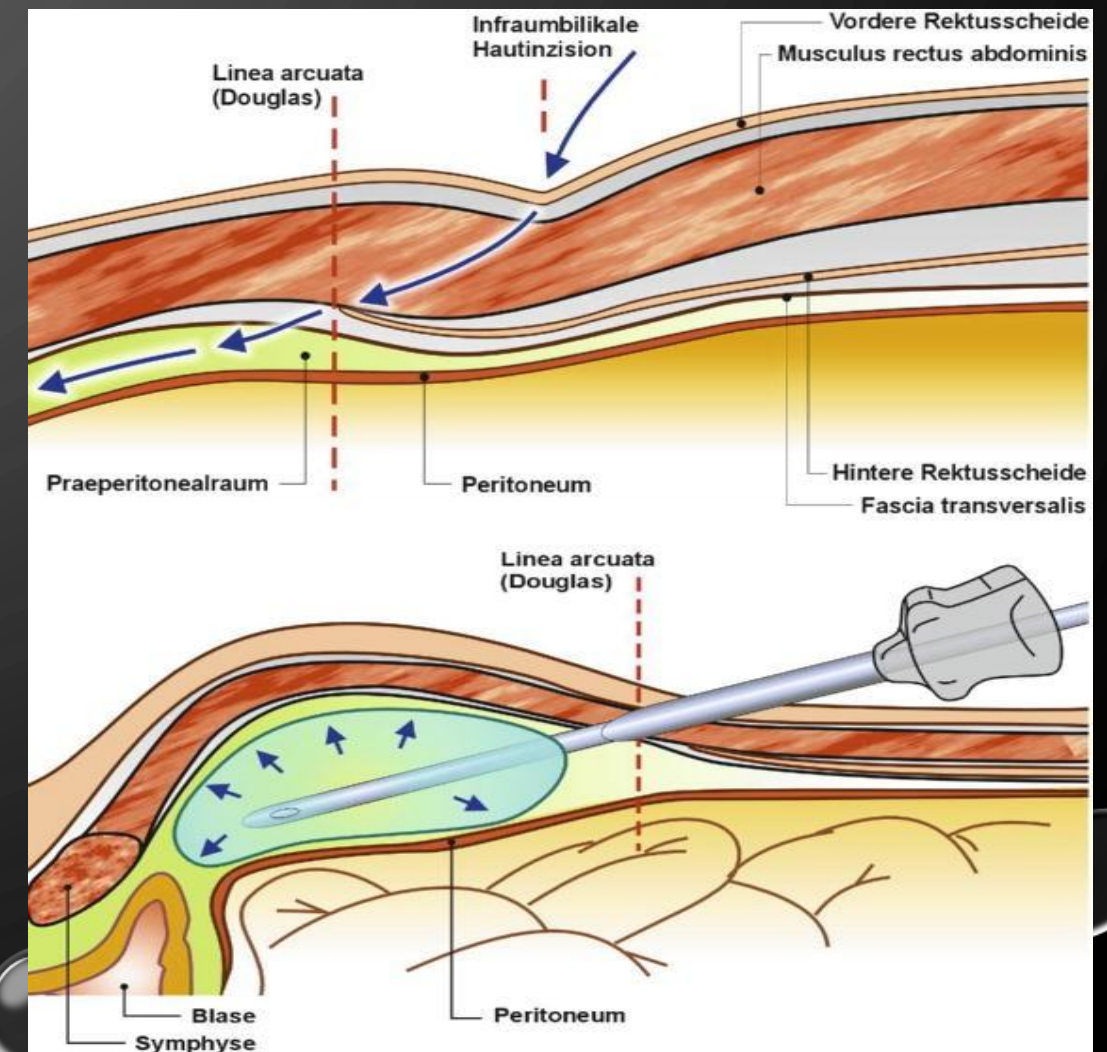
When this condition occurs, it will be found:

: PaCO2 and ETCO2 are higher.

How to ?

: Pause surgery

: When starting a new surgery, start with a small amount of pressure.



Pneumothorax, Pneumomediastinum and Pneumopericardium

It is caused by high intra-abdominal pressure. This allows the gas to move into the chest cavity.

When this condition occurs, it will be found:

: Airway pressure PaCO_2 and ETCO_2 are higher.

How to ?

: Closed nitrous oxide

: Adjust ventilation to correct hypoxemia.

: Use as little pressure in the abdomen as possible for surgery.

: Inform the surgeon to plan the treatment together.

Gas embolism

It is caused by gas entering the circulatory system. This causes a "gas lock" in the aorta and right ventricle.

This causes heart failure and circulatory failure.

Symptoms

: tachycardia, arrhythmia, hypotension, ECG change.

How to ?

: Stop the injection of CO₂ gas immediately and reduce the intra-abdominal pressure.

: Shut down nitrous oxide to prevent gas embolism from enlargement.

: 100% O₂ on and hyperventilation

: Position on the left side of the head low to prevent/reduce gas entering the pulmonary circulation.



Conclusion and Key Takeaways

Pressure injuries and pneumoperitoneum complications are potential risks in perioperative urology nursing.

Understanding risk factors, employing preventive techniques, and adhering to best practices are crucial for patient safety.

Specialized knowledge and skills are paramount in managing these surgical challenges.